

CPU Monitoring

The system-monitoring module is module ID 9. It discovers supported NVIDIA CPUs and their cores and supplies CPU telemetry through the same field-watch model that DCGM uses for GPUs. The module is loaded when system-monitoring functionality is first requested.

When CPU monitoring is useful

CPU monitoring is useful on NVIDIA Grace systems where GPU behavior must be understood together with its host-side workload. It can help answer whether a job is CPU-bound, whether selected cores are saturated, whether CPU power or temperature is approaching a platform limit, and whether CPU clock behavior correlates with a GPU performance change.

The module provides telemetry, not a general-purpose host monitor. It covers the NVIDIA CPU entities and fields that DCGM exposes; it does not replace operating-system tools for processes, memory, storage, or networking. A typical node workflow combines DCGM CPU and GPU fields under the same monitoring system, then uses OS-level tools when a CPU signal needs process-level investigation.

Requirements and entity model

CPU monitoring is supported for NVIDIA Grace CPUs, beginning with DCGM 3.3. On unsupported hosts, CPU discovery is empty and CPU field watches are not applicable.

DCGM uses two entity groups:

- `DCGM_FE_CPU`, selected as `cpu:<id>`, represents a whole CPU.
- `DCGM_FE_CPU_CORE`, selected as `core:<id>`, represents one CPU core.

Temperature and power are CPU-scoped. Utilization and clock fields are core-scoped. Pairing a valid field with the wrong entity type can produce an error or an unavailable value.

Discover CPU and core IDs at runtime:

```
$ dcgmi discovery --list
$ dcgmi discovery --info a --cpuid 0
```

The CPU information report includes the core ranges owned by that CPU. Core IDs should not be inferred from the CPU ID or copied from another host.

CPU and core fields

CPU monitoring fields

Field name	Description	ID	Entity
DCGM_FI_DEV_CPU_UTIL_TOTAL	Total percentage of time in use	1100	Core
DCGM_FI_DEV_CPU_UTIL_USER	Percentage of time used by user processes	1101	Core
DCGM_FI_DEV_CPU_UTIL_NICE	Percentage of time used by low-priority processes	1102	Core
DCGM_FI_DEV_CPU_UTIL_SYS	Percentage of time used by the system	1103	Core
DCGM_FI_DEV_CPU_UTIL_IRQ	Percentage of time used for interrupts	1104	Core
DCGM_FI_DEV_CPU_TEMP_CELSIUS	Current temperature in degrees Celsius	1110	CPU
DCGM_FI_DEV_CPU_CLOCK_CURRENT	Current clock speed in kHz	1120	Core
DCGM_FI_DEV_CPU_POWER_WATTS	Current power use in watts	1130	CPU
DCGM_FI_DEV_CPU_POWER_LIMIT_WATTS	Current power limit in watts	1131	CPU
DCGM_FI_DEV_CPU_VENDOR	CPU vendor name	1140	CPU
DCGM_FI_DEV_CPU_MODEL	CPU model name	1141	CPU

Example: correlate CPU pressure with a workload

Suppose discovery reports CPU 0 and cores 0 through 3 are the cores assigned to a workload. In one terminal, monitor CPU temperature, power, and power limit for ten samples:

```
$ dcgmi dmon --entity-id cpu:0 \  
  --field-id 1110,1130,1131 --count 10
```

In another terminal, monitor the assigned cores at the same one-second cadence:

```
$ dcgmi dmon --entity-id 'core:{0-3}' \  
  --field-id 1100,1101,1103,1120 --delay 1000 --count 10
```

Run the application during the sampling window. Sustained high total utilization on the assigned cores together with lower-than-expected GPU activity can justify investigating a CPU-side bottleneck.

Temperature near a platform limit, power near the reported power limit, and a falling core clock can provide additional context, but none of those signals alone proves the cause of application performance.

For a reusable selection, create a mixed group from the IDs reported by discovery:

```
$ dcgmi group --create job-4815 \  
  --add 'cpu:0,core:{0-3},gpu:0'
```

Not every consuming command accepts every entity type or every mixture. A group is a reusable selector, not a guarantee that one field applies to all of its members. Direct entity selections are often clearer when CPU- and core-scoped fields differ.

Further reading

- [dcgmi discovery](#) documents CPU and core ID discovery.
- [dcgmi dmon](#) documents field selection, cadence, output, and unavailable values.
- [dcgmi group](#) documents entity selectors and persistent groups.
- [Field Identifiers](#), [Field Scope](#), and [Field APIs](#) provide the exhaustive programmatic field reference.



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